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The Estimate Lives in the Keys: What a Transformer Learns When It Adapts to Unknown Physics In-Context

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Abstract

[abstract pending]

Keywords: in-context learning, adaptive control, system identification, dual control, mechanistic interpretability

1. Introduction

A robot picking up an unfamiliar object, or a legged robot stepping onto ice, must adapt to physics it does not know from a few seconds of interaction. A popular recipe trains a sequence model across randomised physics and lets it adapt *in context*, from its own history, with no gradient steps at deployment (Kumar et al., 2021; Duan et al., 2016; Forgione et al., 2023; Busetto et al., 2024). These models work, but it is not clear what they compute. Do they identify the plant and then plan? Do they implement a classical estimator? Do they explore?

For linear-Gaussian dynamics these questions have exact answers, so they can be tested. Transformers trained on random linear systems are known to *behave* like the Kalman filter or least squares (Balim et al., 2023; Akram and Vikalo, 2026). They are also known to be *able to represent* such estimators (Goel and Bartlett, 2024; Cole et al., 2025; Vlădărean et al., 2026). What has not been measured is what is inside a trained in-context system-identifier, or what an in-context *controller* does about the dual effect. The second question matters because, under exploration-free feedback, closed-loop data do not identify the plant (Feldbaum, 1960; Åström and Wittenmark, 1995; Ljung, 1999). Akram and Vikalo (2026) list the first question as future work.

We study both questions in a pre-registered design. Hypotheses, metrics, seeds and tests were committed before the main experiments, and deviations are logged. The design combines exact optimal references, causal interventions and three settings of increasing difficulty: linear prediction, MuJoCo contact physics (Todorov et al., 2012), and adaptive control. Our contributions:

- **Theory (§3).** An exact Bayes-risk identity for in-context prediction under any non-anticipating policy, with a parameter-free context-length law $t\,\mathcal{R}_t^\star \to np\sigma^2$. A closed-loop non-identifiability identity: the information gained along unexcited directions equals the excitation energy, and is exactly zero under pure feedback. A two-layer construction that gathers the exact sufficient statistics, which turns "statistics first, solve later" into a falsifiable probing prediction.
- **What the trained predictor computes (§5). [introA pending]**
- **Contact physics (§6). [introB pending]**
- **In-context control (§7). [introC pending]**

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2. Related work

In-context learning of function classes. Transformers trained on random regression tasks match least squares (Garg et al., 2022). Behaviourally they interpolate between ridge and OLS, and probes recover intermediate quantities in static regression (Akyürek et al., 2023). They can implement GD (von Oswald et al., 2023; Ahn et al., 2023; Zhang et al., 2024), higher-order iterations (Fu et al., 2023) and algorithm selection (Bai et al., 2023), and they approximate posterior predictives (Müller et al., 2022; Raventós et al., 2023). We move these questions to dynamical, closed-loop data, where the regressors depend on the past and on the policy. There the dual effect has no analogue in i.i.d. regression.

In-context learning of dynamical systems. Meta-trained transformers predict and filter unseen linear and nonlinear systems (Forgione et al., 2023; Piga et al., 2023; Busetto et al., 2023; Balim et al., 2023; Akram and Vikalo, 2026) and can serve as contextual controllers for a system class (Busetto et al., 2024). Constructions and approximation theory exist for Kalman filtering (Goel and Bartlett, 2024), for least-squares learning of LDS with depth separation (Cole et al., 2025), and for single linear-attention layers (Vlădărean et al., 2026). All of these are either behavioural or representational. None of them intervenes on a trained model.

Adaptation from history and in-context RL. RMA (Kumar et al., 2021) regresses a history encoding onto privileged physical parameters. Probing such latents gives mixed results (Chamachot, 2026), and decoding them can need a learned generative model (Zhang et al., 2026). Meta-RL agents can learn to explore (Duan et al., 2016; Zintgraf et al., 2020); algorithm distillation imitates learning histories (Laskin et al., 2023); and in-context RL algorithms are provably representable in tabular MDPs (Liang and Lai, 2026). In adaptive LQR, certainty equivalence with dithering is rate-optimal (Mania et al., 2019; Simchowitz and Foster, 2020). We use these results as baselines and ask whether a transformer trained only on cost discovers the excitation they prescribe.

Interpretability tools. We use linear probes (Alain and Bengio, 2016) and activation patching (Zhang and Nanda, 2024; Heimersheim and Nanda, 2024). Our probes are always paired with interventions, and the probe targets are *estimator* quantities, which are distinct from the ground truth.

3. Theory

Setting. A plant $x_{s+1} = \theta z_s + w_s$ with regressor $z_s = (x_s, u_s) \in \mathbb{R}^p$, unknown $\theta = [A\ B] \in \mathbb{R}^{n\times p}$, and noise $w_s \sim \mathcal{N}(0, \sigma^2 I_n)$ i.i.d. and independent of everything else. The rows θ_i of θ have independent priors $\mathcal{N}(0, \Lambda_0^{-1})$. Inputs are chosen by an arbitrary *non-anticipating* policy: u_s is a measurable function of the past data $\mathcal{D}_s = (x_0, u_0, \dots, x_s)$ and of independent randomness, so it does not depend on θ except through $\mathcal{D}_s$. An *in-context predictor* is any map $\hat{y}_t = f_t(\mathcal{D}_t, u_t)$, of which a causal transformer is one example. Write $S_t = \sum_{s<t} z_s z_s^\top$ and $C_t = \sum_{s<t} z_s x_{s+1}^\top$.

Theorem 1 (Exact Bayes risk of in-context prediction) *Under the untruncated Gaussian prior, for every in-context predictor and every t,*

$$\mathbb{E}\,\|\hat{y}_t - \theta z_t\|^2 \;\geq\; \mathcal{R}_t^\star := n\sigma^2\,\mathbb{E}\big[z_t^\top(\sigma^2\Lambda_0 + S_t)^{-1} z_t\big], \tag{1}$$

with equality iff $\hat{y}_t = C_t^\top(\sigma^2\Lambda_0 + S_t)^{-1} z_t$ almost surely. This is ridge regression, i.e. RLS initialised at the prior.

Proof The likelihood of $\mathcal{D}_t$ factorises into the policy factors $p(u_s \mid \mathcal{D}_s)$, which do not depend on θ, and $\prod_{s<t} \mathcal{N}(x_{s+1}; \theta z_s, \sigma^2 I)$. Hence the posterior of each row is Gaussian, $\theta_i \mid \mathcal{D}_t \sim \mathcal{N}(m_i, \Sigma_t)$, the rows are independent, and $\Sigma_t = (\Lambda_0 + \sigma^{-2} S_t)^{-1} = \sigma^2(\sigma^2\Lambda_0 + S_t)^{-1}$ and $[m_1, \dots, m_n] = \Sigma_t \sigma^{-2} C_t$. Since u_t is conditionally independent of θ given $\mathcal{D}_t$, the posterior given $(\mathcal{D}_t, u_t)$ is the same. For any $(\mathcal{D}_t, u_t)$-measurable $\hat{y}_t$, the bias–variance identity gives $\mathbb{E}[\|\hat{y}_t - \theta z_t\|^2 \mid \mathcal{D}_t, u_t] = \|\hat{y}_t - m z_t\|^2 + \sum_i z_t^\top \Sigma_t z_t$. Taking expectations yields (1), with equality iff $\hat{y}_t = m z_t$. ■

Corollary 2 (A parameter-free context-length law) *Suppose that, conditionally on θ, the closed-loop process is ergodic with $S_t/t \to \Sigma_\theta \succ 0$ a.s., the query satisfies $\mathbb{E}[z_t z_t^\top \mid \theta] \to \Sigma_\theta$, and the family $\{t\, z_t^\top (\sigma^2\Lambda_0 + S_t)^{-1} z_t\}_t$ is uniformly integrable. Then $t\,\mathcal{R}_t^\star \to n\,p\,\sigma^2$. The limit does not depend on the prior, on θ, or on the input distribution. The rate also holds for open-loop excitation of any stable plant (as in Experiment A), because* $\mathrm{tr}(\Sigma_\theta^{-1}\Sigma_\theta) = p$ *whatever Σ_θ is.*

Proof $t(\sigma^2\Lambda_0 + S_t)^{-1} \to \Sigma_\theta^{-1}$ a.s. Uniform integrability lets us exchange the limit and the expectation, which gives $\mathbb{E}[t\, z_t^\top (\sigma^2\Lambda_0 + S_t)^{-1} z_t \mid \theta] \to \mathrm{tr}(\Sigma_\theta^{-1}\Sigma_\theta) = p$. Multiply by $n\sigma^2$ and integrate over θ. ■

Theorem 1 turns "is the transformer Bayes-optimal?" into a measurable gap. Corollary 2 gives a target curve $\mathcal{R}_t^\star \approx np\sigma^2/t$ that involves no fitted constants. (In Experiment A, the prior is truncated to stable A. The exact optimum is then the truncated posterior mean, which we compute by Monte Carlo. The truncation only matters at short contexts.)

Proposition 3 (Closed-loop non-identifiability; the dual effect) *Let the policy be $u_s = Kx_s + \varepsilon_s$ for a fixed $K \in \mathbb{R}^{m\times n}$ and an excitation sequence ε_s. For any $b \in \mathbb{R}^m$, let $v_b = (-K^\top b,\ b) \in \mathbb{R}^p$. Then for all t,*

$$v_b^\top \Sigma_t^{-1} v_b \;=\; v_b^\top \Lambda_0 v_b + \sigma^{-2} \textstyle\sum_{s<t} (b^\top \varepsilon_s)^2. \tag{2}$$

In particular, without excitation ($\varepsilon \equiv 0$), the posterior precision of every row along v_b stays at its prior value for every context length. By Cauchy–Schwarz, the posterior variance $v_b^\top \Sigma_t v_b \geq \|v_b\|^4/(v_b^\top \Lambda_0 v_b)$ never shrinks below the prior level. So the combination $\theta v_b = (B - AK^\top)b$ is not learned, and no in-context learner can do better than the prior along it.

Proof $v_b^\top z_s = -b^\top K x_s + b^\top(Kx_s + \varepsilon_s) = b^\top \varepsilon_s$, so $v_b^\top S_t v_b = \sum_{s<t} (b^\top \varepsilon_s)^2$. Substitute into $\Sigma_t^{-1} = \Lambda_0 + \sigma^{-2} S_t$. The variance bound follows from $(v^\top v)^2 \leq (v^\top \Sigma v)(v^\top \Sigma^{-1} v)$. ■

The extreme case is $K = 0$ with a prior mean of zero for B. A certainty-equivalent controller then outputs $u \equiv 0$ forever and never learns anything about B; Experiment C shows this failure exactly. A good adaptive controller must therefore *spend* control effort on excitation. The information it gathers along the unexcited directions equals the energy of that excitation, which we measure as $\log\det(\Lambda_0 + \sigma^{-2} S_8)$.

Proposition 4 (Two attention layers suffice to gather the sufficient statistics) *Let token t be $e_t = (z_t, 1, \pi_t)$, with $\pi_t \in \{0,1\}^T$ a one-hot position. There is a two-layer causal* linear-*attention network (one head in layer 1, p heads in layer 2, no MLPs) whose residual stream at every position t contains S_t and C_t* exactly*, together with z_t and t. Consequently, the Bayes predictor of Theorem 1 is a fixed continuous function $\phi(S_t, C_t, z_t)$ of the token-t residual alone. A token-wise MLP approximates it uniformly on any compact set, and every additional layer can only serve the "solve" ϕ, not the gathering.*

Proof *Layer 1 (previous-token head).* Set $q_t = \Pi\pi_t = \pi_{t-1}$, with Π the down-shift and $\pi_{-1} = 0$; set $k_s = \pi_s$ and $v_s = (z_s, x_s)$, where x_s is read off z_s. Linear attention outputs $\sum_{s\le t} v_s\, k_s^\top q_t = v_{t-1}$. The residual becomes $(z_t, 1, \pi_t, z_{t-1})$, with $z_{-1} = 0$. *Layer 2 (moment heads).* Head $j \in \{1..p\}$ uses the constant query $q_t = 1$, the key $k_s = (z_{s-1})_j$, and the value $v_s = (z_{s-1}, x_s)$. It outputs $\sum_{s\le t}(z_{s-1}, x_s)(z_{s-1})_j$, which is the j-th column of $[S_t; C_t^\top]$, because the pairs (z_{s-1}, x_s), $1 \le s \le t$, are exactly the context pairs. Writing the p head outputs to disjoint residual coordinates gives S_t and C_t. The map ϕ is continuous on $\{\sigma^2\Lambda_0 + S \succ 0\}$, so the universal approximation theorem applies on compacta. ■

Remark 5 (Where an estimate can live without appearing in any activation) *Consider a softmax head with keys k_s, values v_s and query q, and suppose its scores are small, $|q^\top k_s|/\sqrt{d_h} \ll 1$. This is the near-uniform regime, with normalised attention entropy close to 1. Expanding the softmax to first order,*

$$\sum_s \mathrm{softmax}_s\, v_s \;\approx\; \bar{v} + \tfrac{1}{\sqrt{d_h}}\,\widehat{\mathrm{Cov}}(v,k)\,q, \qquad \widehat{\mathrm{Cov}}(v,k) = \tfrac{1}{t+1}\textstyle\sum_{s\le t}(v_s - \bar{v})k_s^\top. \tag{3}$$

The head is therefore an affine map of the query, and its matrix is the key–value covariance ("fast weights" (von Oswald et al., 2023)), which is spread over all context positions. A transformer can thus apply a parameter estimate to the query, with the predictor's Jacobian $\partial\hat{y}_t/\partial z_t$ playing the role of $\hat{\theta}_t$, while $\hat{\theta}_t$ never appears as a vector in the residual stream. Two predictions follow. A residual probe for $\hat{\theta}$ can fail even when the model is near Bayes-optimal; a probe on $\widehat{\mathrm{Cov}}(v,k)$, and patching of keys and values alone, should succeed. Section 5 confirms both.

Proposition 4 is in the spirit of earlier constructions (Akyürek et al., 2023; Bai et al., 2023; von Oswald et al., 2023), and we do not claim it as a contribution in itself. Its role is to make a *falsifiable* mechanistic prediction for trained models. If a trained transformer works this way, a linear probe should decode the normalised statistics $(S_t, C_t)/(t{+}1)$ no later in depth than it decodes the posterior mean $\hat{\theta}_t$ (H2c). If it instead refines $\hat{\theta}$ iteratively in the style of GD (von Oswald et al., 2023), $\hat{\theta}$ should become decodable gradually, with no early peak for the statistics.

4. Experimental design

Pre-registration. Before the main experiments we committed the hypotheses, metrics, seeds, thresholds and analysis plan (`PREREGISTRATION.md`, git commit `d134c5a`). Deviations are listed in `docs/DEVIATIONS.md` and in Appendix A. All confirmatory claims use a hierarchical bootstrap (training seeds, then test episodes; 10^4 resamples) with Holm–Bonferroni correction across the 13 confirmatory tests.

Model. Every experiment uses the same GPT-style causal decoder: pre-LN, d=128, 4 layers, 4 heads, MLP width 512, learned positions, about 0.8M parameters. Each model is trained from scratch with AdamW and three seeds, on a 4 GB laptop GPU. "Residual stream ℓ" means the input to block $\ell{+}1$ (ℓ=0: embeddings).

A: in-context prediction of linear systems. $x \in \mathbb{R}^2$, $u \in \mathbb{R}$ (p=3), σ=0.1. The prior is $A_{ij} \sim \mathcal{N}(0, 0.4^2)$, $B_i \sim \mathcal{N}(0, 0.5^2)$, truncated to $\rho(A) < 0.95$, with $u_t \sim \mathcal{N}(0,1)$ and T=48. Token t is (x_t, u_t) and the target is x_{t+1}. Batches are freshly sampled for 20k steps. References: the *exact*

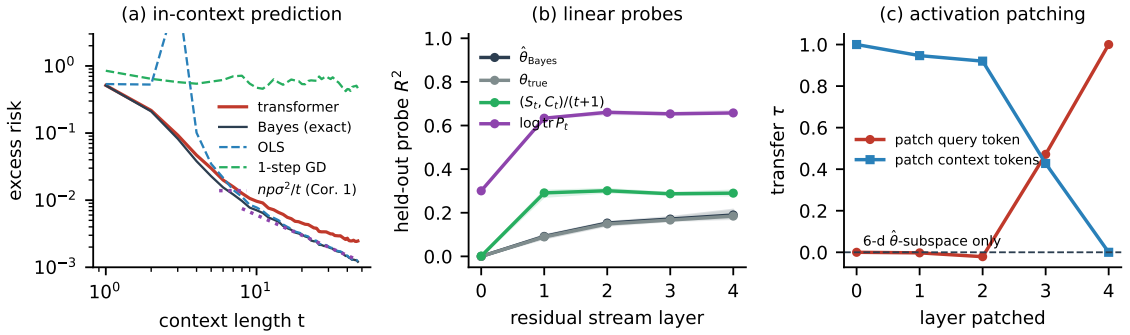


Figure 1: **A, pre-registered analyses** (3 seeds). (a) Excess risk vs. context length. The transformer tracks the exact Bayes predictor while data are scarce, then flattens at an approximation floor. The Bayes curve follows the parameter-free law of Cor. 2. (b) Held-out linear probes of the residual stream (band: range over seeds). The estimate $\hat{\theta}_t$ is barely decodable; the posterior uncertainty is. (c) Activation patching of episode A into B. Patching the query token transfers A's behaviour only from layer 3 on. Patching the 6-d $\hat{\theta}$-probe subspace transfers nothing (dashed).

Bayes predictor (truncated posterior mean, Monte Carlo with 2048 samples), ridge/RLS, OLS, one-step GD with tuned step size (von Oswald et al., 2023), and zero. Excess risk is measured against the noiseless mean θz_t.

B: MuJoCo physics. *Pendulum*: point mass on a 0.5 m rod, with $m \sim U[0.5, 2]$, joint damping $c \sim U[0.02, 0.3]$, $g \sim U[4, 16]$, random torques. Feature regression on $(u, \dot{q}, \sin q, 1)$ contains the true model class. *Pushed block*: a box on a plane with Coulomb contact friction $\mu \sim U[0.1, 1]$, $m \sim U[0.5, 3]$, and random forces $F \sim U[-15, 15]$ N. About half of the steps are stuck, so the stick–slip dynamics fall outside any linear-in-parameters model. Both systems use control step 0.05 s, unobserved process noise, 60k training episodes and T=48. Baselines: simulation with the true parameters, simulation with nominal parameters, physics-feature RLS, and a no-context MLP.

C: in-context adaptive control. $A = 0.9I + \mathcal{N}(0, 0.25^2)$ truncated to $\rho(A) < 1.1$, so some plants are open-loop unstable. $B = r(\cos\phi, \sin\phi)$ with ϕ uniform, so the sign and direction of actuation are unknown. Cost $J = \sum_{t<48} \|x_t\|^2 + 0.1u_t^2 + \|x_{48}\|^2$, $|u| \le 5$. The transformer reads (x_t, u_{t-1}), outputs u_t, and is trained by back-propagating J through the differentiable plant. It never sees θ and has no system-ID loss. Baselines, all tuned on separate plants: oracle finite-horizon LQR; the best constant gain over the prior ("fixed"); certainty equivalence (CE: ridge estimate, then LQR on the remaining horizon); CE with decaying dither $s_0/\sqrt{t+1}$ (Simchowitz and Foster, 2020); and explore-then-CE (system ID followed by control). Test plants and noise are shared across controllers.

5. Experiment A: what the in-context predictor computes

Behaviour: Bayes-optimal while data are scarce, then an approximation floor. The trained predictor is much closer to the exact Bayes predictor than to OLS or to one GD step. The mean algorithmic distance to OLS exceeds the distance to Bayes by 0.271 [0.181, 0.379], and the distance

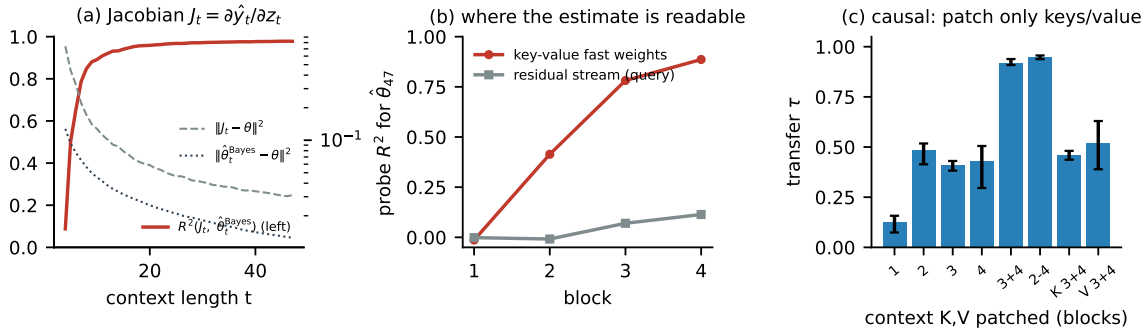


Figure 2: **A, exploratory follow-up: where the estimate lives** (3 seeds). (a) The predictor's Jacobian with respect to its query, $J_t{=}\partial\hat{y}_t/\partial z_t$, matches the Bayes estimate. Its error to the truth (dashed) tracks the Bayes estimate's error (dotted). (b) $\hat{\theta}_{47}$ is linearly readable from the per-block key–value fast weights $\widehat{\mathrm{Cov}}(v,k)$ but not from the residual stream at the same position. (c) Patching *only* the context keys and values of blocks 3–4, leaving every activation of the query untouched, transfers over 90% of the other episode's behaviour. Keys or values alone transfer about half.

to GD exceeds it by 0.552 [0.496, 0.616] (H1a, H1b; Holm $p < 10^{-3}$, $< 10^{-3}$). For $t \leq 4$ its excess risk is within a factor 1.23 of Bayes. After that it decays more slowly than $1/t$ and settles about 1.3×10^{-3} above Bayes. Meanwhile $t\,\mathcal{R}_t^\star$ reaches 0.057 at $t{=}47$: Theorem 1 gives 0.058 and Corollary 2 predicts $np\sigma^2{=}0.06$. The pre-registered tolerance H1c is **rejected**: the mean excess ratio is 1.59 [1.56, 1.62], against a threshold of 1.25. This is the dynamical analogue of the observation that trained transformers fall short of least-squares rates (Hill et al., 2025; Cole et al., 2025). Longer training and larger models narrow the gap but do not close it (§8).

The pre-registered mechanism is rejected: no estimate in the residual stream. The best linear probe of the residual stream for $\hat{\theta}_t$ reaches only $R^2{=}0.19$ [0.17, 0.21] (H2a, threshold 0.8: rejected). It beats the probe for the true θ by just 0.005 (H2b: significant but negligible). The normalised sufficient statistics are partly decodable ($R^2{\leq}0.30$) and peak at layers 1–2, before $\hat{\theta}$ peaks at layer 4, in every seed. So H2c holds formally but is uninformative. What *is* linearly decodable is the model's uncertainty: $\log \mathrm{tr}\, P_t$ reaches $R^2{=}0.66$. Steering along the $\hat{\theta}$-probe directions does not move the prediction (slope -0.0000; H3a rejected). Swapping the 6-d probe subspace of the query between episodes transfers $\tau{=}0.0001$ of the behaviour, which is less than a random 6-d subspace (0.067; H3b rejected). Whole-stream patching (Fig. 1c) shows that parameter information enters the query position only in blocks 3–4: $\tau_{\text{last}}{=}0.38 - -0.62$ after block 3, while patching all context positions at layer 2 still transfers ≥ 0.91.

Exploratory: the estimate lives in the keys and values, and acts through the query Jacobian. The following analyses were added after the pre-registered ones failed, and they are labelled as such. Remark 5 suggests where to look.

- *The answer is linearly represented; the parameter is not.* A linear probe recovers the Bayes *prediction* $\hat{\theta}_t z_t$ with $R^2{=}0.88 - -0.92$ after block 3 and $0.97 - -0.98$ after block 4. An MLP probe recovers $\hat{\theta}_t$ itself only up to $R^2{=}0.72$.

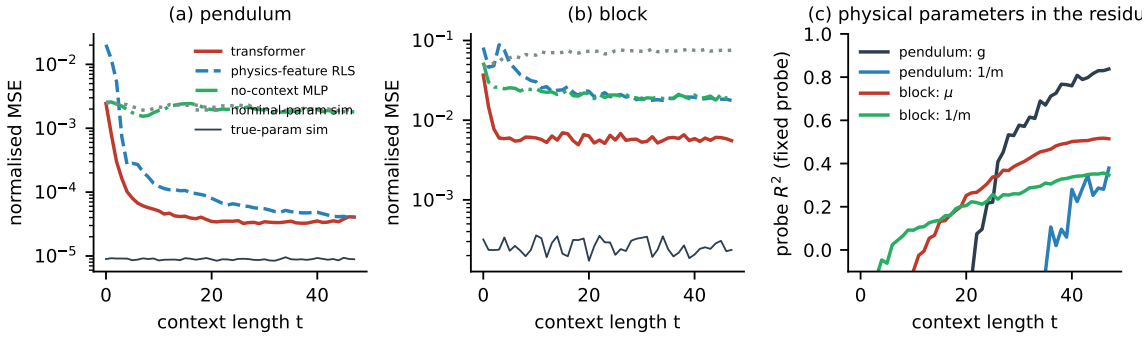


Figure 3: **B** (3 seeds). (a, b) Normalised one-step error vs. context length. On the pendulum the transformer beats physics-feature RLS, even though RLS has the true model class up to discretisation. On the stick–slip block it beats RLS by $3\times$. (c) Residual-stream probes (fixed final-position probe applied at every t).

- *The Jacobian is the estimate.* The predictor is close to linear in its own query: counterfactual queries are explained with $R^2{=}0.87-{-}0.96$ by its Jacobian, and their effect has slope 0.97–0.99 against the Bayes map. Its Jacobian $J_t{=}\partial\hat{y}_t/\partial z_t$ matches $\hat{\theta}_t^{\text{Bayes}}$ with $R^2{=}0.92-{-}0.92$, more closely than it matches the truth $(0.89-{-}0.89)$ or OLS $(0.90-{-}0.90)$ (Fig. 2a).

- *It is stored in key–value memory.* The per-head fast weights $\widehat{\text{Cov}}(v,k)$ of block 4 predict $\hat{\theta}_{47}$ with $R^2{=}0.88-{-}0.89$ (block 3: 0.78–0.79). The residual stream at the same position gives at most 0.08–0.14 (Fig. 2b).

- *The memory is causal.* Replacing only the context keys and values of blocks 3–4 by those of another episode, with the query's residual stream, its attention queries and all MLPs left as they were, transfers $\tau{=}0.91-{-}0.94$ of that episode's behaviour (blocks 2–4: 0.94–0.96). Keys alone give 0.44–0.48 and values alone 0.39–0.63(Fig. 2c). The carriers are 2–3 heads per block with near-uniform attention, as in Remark 5.

- *Probe directions are not the model's directions.* Even for the well-decoded prediction, steering along the probe direction has slope only 0.08, and swapping the probe subspace transfers only 0.08. Linear decodability did not imply causal use anywhere in this study.

In short, the transformer does implement near-Bayes system identification. It does not do so by writing $\hat{\theta}$ into an activation. The estimate is spread over the key–value memory of the context and is read out, applied to the query, by attention. This is why RMA-style "latent parameter" probes of the residual stream (Kumar et al., 2021; Chamachot, 2026) can fail even when adaptation works.

6. Experiment B: nonlinear and contact physics in MuJoCo

Prediction. Averaged over $t{=}24..47$, the normalised error on the pushed block with Coulomb friction is 0.0058 for the transformer and 0.019 for physics-feature RLS (H4a: difference 0.0131 [0.0128, 0.0134], Holm $p < 10^{-3}$). For comparison, the no-context MLP gets 0.019, nominal-parameter simulation 0.075, and simulation with the true parameters 0.00026. On the pendulum,

where feature RLS has the right model class up to discretisation, the transformer (3.5e-05) still beats it (5e-05). Nominal simulation (0.002) and the no-context MLP (0.0019) are $\sim$60$\times$ worse, so the gain comes from in-context adaptation.

Mechanism. The pre-registered probes again fail for the parameters that enter the dynamics multiplicatively. Residual-stream R^2 at t=47 is 0.82–0.86 for g, 0.32–0.44 for $1/m$ and 0.51–0.52 for μ (H4c rejected). Steering along the $1/m$ probe moves the input sensitivity the right way in only 42% (pendulum) and 71% (block) of episodes (H4d rejected). The exploratory analyses of §5 transfer. The model's own sensitivity $\partial \dot{q}_{t+1}/\partial u_t$, its functional estimate of $1/m$, correlates with the true $1/m$ at r=0.99 – −0.99 on the pendulum; on the block it is r=0.42 – −0.43, because the sensitivity is zero while stuck. The key–value fast weights decode $1/m$ with R^2=nan – −nan (residual: nan–nan), and on the block they decode μ at nan–nan (residual nan–nan). **[bouncer pending]**

7. Experiment C: does an in-context controller explore?

[resC pending]

8. Scale and training length

[resScale pending]

9. Discussion and limitations

[discussion pending]

Generative AI use

Generative AI tools (Claude) were used for literature search and synthesis, code implementation, mathematical derivations, and writing. The author reviewed the content, every reported number is regenerated by the scripts in the public repository, and the author takes responsibility for the final content.

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Appendix A. Deviations from the pre-registration

(1) We measure excess risk against the noiseless mean θz_t instead of as MSE $- n\sigma^2$. The two have the same expectation and ours has lower variance. (2) $\Lambda_0{=}I$ in the information measure, which the pre-registration did not fix. (3) Training details not fixed by the pre-registration: the constant-gain baseline uses Adam, 1000 steps and batch 1024; the no-context MLP has $2{\times}256$ units and 5000 steps. (4) The controller uses a KV-cached incremental forward to fit in 4 GB. It is numerically identical to the full forward. (5) With a prior mean $B{=}0$, pre-registered CE degenerates to $u \equiv 0$; we report it as specified. (6) On the pendulum the parameter effects are small (nominal-parameter simulation already has 0.5% normalised error); we kept the design unchanged. **[devextra pending]**

Appendix B. Implementation details

Exact Bayes predictor (A). Each row of θ has the Gaussian posterior of Theorem 1, and the rows share their covariance. We draw 2048 joint samples, discard those with $\rho(A) \geq 0.95$, and average. The acceptance rate is **[accz pending]** at $t{=}0$ and **[accl pending]** at $t{=}47$. **Probes.** Ridge regression on standardised residuals, with α chosen by 5-fold CV grouped by episode, fitted on 2048 test episodes and scored on the other 2048. **Steering.** Let D be the probe readout matrix. The minimal-norm residual change that moves readout coordinate k by δ while leaving the others unchanged is $\delta\,(D^{+})_k$. **Patching transfer** is $\tau = \langle \hat{y}_{\text{patch}} - \hat{y}_B, \hat{y}_{A'} - \hat{y}_B \rangle / \|\hat{y}_{A'} - \hat{y}_B\|^2$, where A' is episode A's context followed by episode B's query token. **Compute.** Everything runs on a laptop RTX 3050 Ti (4 GB) with a 20-thread CPU, in about **[hours pending]** hours end to end.

[appextra pending]